Seminar: Reproducibility of Scientific Results in Software Engineering Research

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LMU LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

SoSy-Lab Software Systems

Warm-up: Who all is here?

Quick round of self-introductions:

- What is your name?
- Do you have any experiences reproducing research results?
- What are your expectations?

Quality Attributes of Research Results

1. Reliability of results
   - Study repeated $\rightarrow$ same conclusions?
   - If not $\rightarrow$ result may be by mere coincidence
2. Applicability of results
   - Availability of prototypes, study instruments, etc.?
   - Usability of prototypes, study instruments, etc.?
   - Transferability or generalizability of conclusions to other settings?

Different Levels of Research Reliability

Repeatability The measurement can be obtained with stated precision by the **same team** using the **same measurement procedure**, the **same measuring system**, under the **same operating conditions**, in the **same location** on multiple trials. For computational experiments, this means that a researcher can reliably repeat her own computation.

Reproducibility The measurement can be obtained with stated precision by a **different team** using the same measurement procedure, the same measuring system, under the same operating conditions, in the same or a different location on multiple trials. For computational experiments, this means that an independent group can obtain the same result using the author's own artifacts.

Replicability The measurement can be obtained with stated precision by a **different team**, a **different measuring system**, in a **different location** on multiple trials. For computational experiments, this means that an independent group can obtain the same result using artifacts which they develop completely independently.

https://www.acm.org/publications/policies/artifact-review-and-badging-current

The Reproducibility Crisis in Science

- 2016: $>70\,\%$ of 1576 scientists unable to reproduce peers' results (https://doi.org/10.1038/533452a)

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- Many follow-up studies, mainly focused on medical and social sciences

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- How about CS?

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- Many follow-up studies, mainly focused on medical and social sciences
- How about CS?
- NASEM report 2019: Root cause for non-reproducibility often lies in *artifact* deficiencies (https://doi.org/10.17226/25303)

Artifacts in Software Engineering Research

Artifact: "a digital object that was either created by the authors to be used as part of the study or generated by the experiment itself"
https://www.acm.org/publications/policies/artifact-review-and-badging-current

Examples:

- Tools
- Scripts to run experiments
- Data (raw or processed)
- Documentation
- Mathematical proof (manual or automated)
- Audio and video materials

There can be *much more* out there!

Systematic Assessment of Artifacts: Artifact Evaluations

- Pioneered at FSE 2011 & OOPSLA 2013
- Initial criteria (https://artifact-eval.org/about.html)
  - consistent with the paper
  - as complete as possible
  - well documented
  - easy to reuse, facilitating further research

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  - easy to reuse, facilitating further research
  - Problem: *Prototypes* maybe not easy to reuse, but useful for reproducibility



Evolution of Artifact Badges

Orthogonal aspects:

- Reusability

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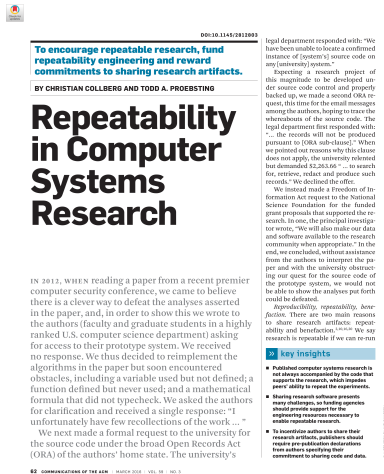
Orthogonal aspects:

- Reusability
- Availability
  - Intellectual property, licensing
  - Security
  - Privacy

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contributed articles

DOI:10.1145/2812803

To encourage repeatable research, fund repeatability engineering and reward commitments to sharing research artifacts.

BY CHRISTIAN COLLBERG AND TODD A. PROEBSTING

Repeatability in Computer Systems Research

IN 2012, WHEN reading a paper from a recent premier computer security conference, we came to believe there is a clever way to defeat the analyses asserted in the paper, and, in order to show this we wrote to the authors (faculty and graduate students in a highly ranked U.S. computer science department) asking for access to their prototype system. We received no response. We thus decided to reimplement the algorithms in the paper but soon encountered obstacles, including a variable used but not defined; a function defined but never used; and a mathematical formula that did not typecheck. We asked the authors for clarification and received a single response: "I unfortunately have few recollections of the work ... "

We next made a formal request to the university for the source code under the broad Open Records Act (ORA) of the authors' home state. The university's legal department responded with: "We have been unable to locate a confirmed instance of [system's] source code on any [university] system."

Expecting a research project of this magnitude to be developed under source code control and properly backed up, we made a second ORA request, this time for the email messages among the authors, hoping to trace the whereabouts of the source code. The legal department first responded with: "... the records will not be produced pursuant to [ORA sub-clause]." When we pointed out reasons why this clause does not apply, the university relented but demanded $2,263.66 "... to search for, retrieve, redact and produce such records." We declined the offer.

We instead made a Freedom of Information Act request to the National Science Foundation for the funded grant proposals that supported the research. In one, the principal investigator wrote, "We will also make our data and software available to the research community when appropriate." In the end, we concluded, without assistance from the authors to interpret the paper and with the university obstructing our quest for the source code of the prototype system, we would not be able to show the analyses put forth could be defeated.

Reproducibility, repeatability, benefaction. There are two main reasons to share research artifacts: repeatability and benefaction.[19,6,8] We say research is repeatable if we can re-run

key insights

- **Published computer systems research is not always accompanied by the code that supports the research, which impedes peers' ability to repeat the experiments.**
- **Sharing research software presents many challenges, so funding agencies should provide support for the engineering resources necessary to enable repeatable research.**
- **To incentivize authors to share their research artifacts, publishers should require pre-publication declarations from authors specifying their commitment to sharing code and data.**

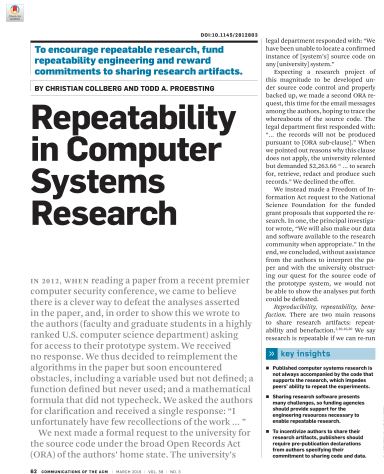
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  - Intellectual property, licensing
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→ Separation of concerns



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Artifact Evaluation – Revised Badges

- ACM initiative 2017:

  https://www.acm.org/publications/task-force-on-data-software-and-reproducibility

- Minor revision 2020 for compliance with NISO RP-31-2021:

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Does the artifact work?

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Is the artifact permanently available?

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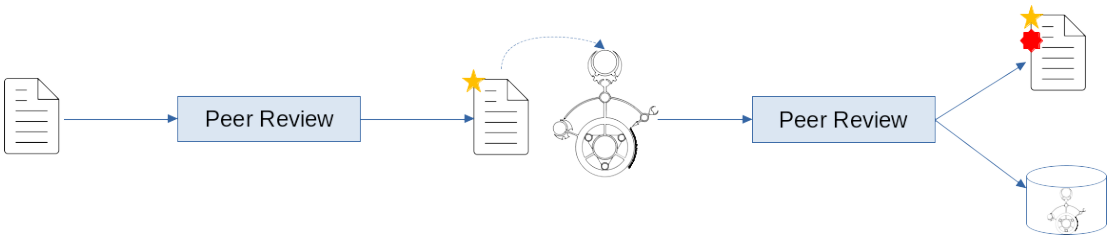
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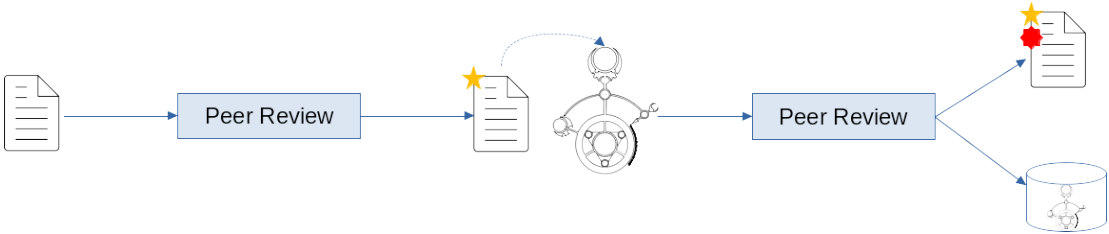


Can the results be confirmed?

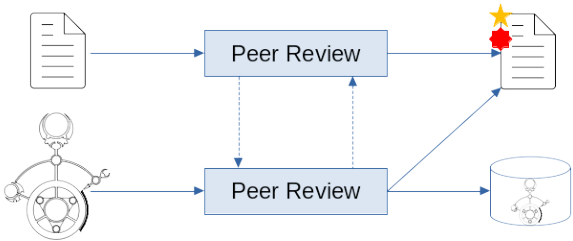
Artifact Evaluation – Processes



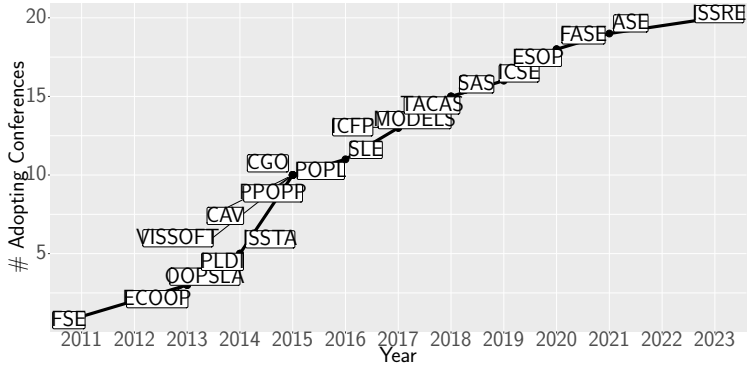
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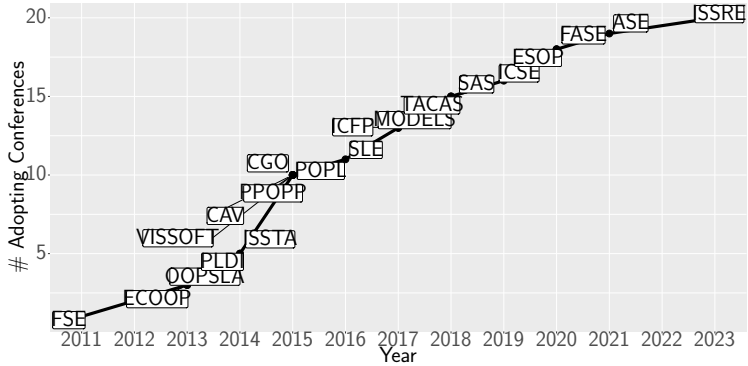
TACAS, ~~OOPSLA~~, ~~ECOOP~~:



Artifact Evaluation – Adoption in Software Engineering



Artifact Evaluation – Adoption in Software Engineering



Everybody does AE. Have we reached the peak of research excellence?

Does AE Solve the Reproducibility Crisis?

Does AE Solve the Reproducibility Crisis?

Thank you for contributing to an answer in this seminar.

Seminar Structure

The seminar comprises 3 phases

1. Read/Discuss background literature (4 weeks, through Nov 10)
2. *Reproduce* research results (6 weeks, through Dec 22, including example-driven tutorial on Nov 17)
3. Consolidate & present results (1 week, through Jan 12)

Introductory Phase: Readings on Reproducibility in SE

- 4 seminar sessions (Tue, 10 AM c.t.)
- Two articles per seminar session
- Read articles (see next slide) in advance
- Each session moderated by 4 participants
  - Initiate discussion (e.g., by providing a short initial summary)
  - Query others' opinions (e.g. by asking for unclear aspects or opinions on central points of the articles)
  - Keep a rough protocol of the discussion (who participates, what are central points discussed)

Introductory Phase: Readings on Reproducibility in SE (Ctd.)

Oct 20 Christian Collberg, Todd A. Proebsting: "Repeatability in Computer Systems Research" (https://doi.org/10.1145/2812803)

Robert Heumüller, Sebastian Nielebock, Jacob Krüger, Frank Ortmeier: "Publish or perish, but do not forget your software artifacts" (https://doi.org/10.1007/s10664-020-09851-6)

Oct 27 Ben Hermann, Stefan Winter, Janet Siegmund: "Community expectations for research artifacts and evaluation processes" (https://doi.org/10.1145/3368089.3409767)

Christopher S. Timperley, Lauren Herckis, Claire Le Goues, and Michael Hilton: "Understanding and improving artifact sharing in software engineering research" (https://doi.org/10.1007/s10664-021-09973-5)

Introductory Phase: Readings on Reproducibility in SE (Ctd.)

Nov 3 Stefan Winter, Christopher S. Timperley, Ben Hermann, Jürgen Cito, Jonathan Bell, Michael Hilton, and Dirk Beyer: "A retrospective study of one decade of artifact evaluations" (https://doi.org/10.1145/3540250.3549172)

Daniel Olszewski, Allison Lu, Carson Stillman, Kevin Warren, Cole Kitroser, Alejandro Pascual, Divyajyoti Ukirde, Kevin Butler, and Patrick Traynor "Get in Researchers; We're Measuring Reproducibility: A Reproducibility Study of Machine Learning Papers in Tier 1 Security Conferences" (https://doi.org/10.1145/3576915.3623130)

Nov 10 Cathy Guevara-Vega, Beatriz Bernárdez, Margarita Cruz, Amador Durán, Antonio Ruiz-Cortés, Martin Solari: "Research artifacts for human-oriented experiments in software engineering: An ACM badges-driven structure proposal" (https://doi.org/10.1016/j.jss.2024.112187)

David Broneske, Sebastian Kittan, and Jacob Krüger: "Sharing Software-Evolution Datasets: Practices, Challenges, and Recommendations" (https://doi.org/10.1145/3660798)

Artifact Analysis Phase

- 5 articles per participant (plus one bonus article for grade improvement)
- Each article is accompanied with a *research artifact* for reproducing results
- To be analyzed over 6 weeks
- Analysis guidelines
  - Call for artifacts (general expectations on artifact format, etc.)
  - Artifact documentation (setup, usage, if you're lucky: detailed steps for reproducing results)
  - Lab report template (how to structure the analysis and report its results)

Artifact Analysis Phase (Ctd.)

- Article assignment in CW 45
- Assignment = Paper DOIs
- By[1] CW 45: E-mail us if the artifact is not available for download or you cannot fulfill its hardware requirements.
  We can only help you if we know early enough.
  Later requests may not be accommodated and the report graded 5.0.

[1] i.e., up until the end of the CW

Artifact Analysis Phase (Ctd.)

- Max. analysis time **(not run time)** per artifact: 8h
- No result after 8h of active work → report why evaluation failed
- **Please track your time**
- Lab report submission system:
  https://gitlab2.cip.ifi.lmu.de/sosy-lab/2026-ss-reproducibility
- Global repository for all papers analyzed in the seminar
- Interactions via issue tracker and merge requests
  - We help you if you face problems
  - You can help each other (& get a bonus on your grade)
  - **But we expect you to make an effort yourselves and Issues should reflect that**

Artifact Analysis Phase (Ctd.)

- First lab report submission deadline (3 reports): Dec 1
- Second lab report submission deadline (2-3 reports): Dec 22
- "Submission": *Feature-complete* – All sections in lab report completed and concluded with a verdict, raw data from reproducibility attempt included.

Result Consolidation & Presentation Phase

- Seminar report: Summary of lab reports + discussion of lessons learned
- Result presentation in block seminar
  - 10 minutes experience report per student
  - 5 minutes discussion
  - **Tentative date: Jan 19, one or two blocks**

Seminar Deliverables

- Literature presentation and discussion (20%)
- 5 lab reports, 1 per analyzed artifact (50%), grade bonus for extra report
- Final report (15%)
  - summarizing the findings from the artifact analysis
  - concluding what you learned
- Result presentation (15%)

Helping others with Issues can improve your grade.

Other Expectations

- Pro-active participation: Engage in discussions
- Continuous participation: Don't lose track
- Self-reliant workload management

Communication

- Zulip channel:
  https://chat.ifi.lmu.de/#narrow/channel/554-Reproducibility-Seminar-WS-2026/
- Gitlab: TBA
- In person: Tue, 10:15 Ludwigstr. 28, room 026